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Bhuvann Vinodh Ram

Data and AI Graduate · MSc AI for Medicine (UCD)

Stamp 1G holder: eligible to work full-time in Ireland without a work permit

PROFILE

MSc graduate in Artificial Intelligence for Medicine from University College Dublin with hands-on experience across the data lifecycle: building pipelines from public data sources, analysing and visualising data, and training, explaining and deploying machine learning models for imaging, signals, text and genomics. Every project below has public code, and most have tests, CI and a live demo. Currently a Trust and Safety Associate at Accenture, Dublin.

EDUCATION

MSc Artificial Intelligence for Medicine and Medical Research — 2024 – 2025
University College Dublin — GPA 3.37 / 4.2

BTech Computer Science and Medical Engineering — 2020 – 2024
Sri Ramachandra Institute of Higher Education and Research, Chennai — CGPA 3.18 / 4

TECHNICAL SKILLS

Languages	Python, SQL, R, MATLAB (basic), JavaScript, Bash
Machine learning	scikit-learn, XGBoost, TensorFlow/Keras, PyTorch; CNNs, BiLSTM, VAEs, U-Net; SHAP
Data	pandas, NumPy, SciPy, ETL pipelines, REST APIs, web scraping, statistics, survival analysis
Vision	OpenCV, OpenCV.js, Detectron2, Tesseract.js
Visualisation	Tableau, matplotlib, seaborn, Plotly
Engineering	Git, GitHub Actions, GitLab CI/CD, Docker, pytest, Streamlit, Gradio, MySQL, MongoDB (basic)

EXPERIENCE

Trust and Safety Associate — Mar 2026 – Present
Accenture — Dublin, Ireland

- Evaluate, classify and moderate user-generated content against client platform policies to keep online communities safe and compliant.

Research Intern — Apr 2025 – Sep 2025
University College Dublin — Dublin, Ireland

- Integrated TCGA colon adenocarcinoma RNA-seq, copy-number and DNA methylation data and trained a variational autoencoder to learn a shared latent representation.
- Clustered the latent space (K-Means, HDBSCAN), evaluated with silhouette scores, and tested clinical relevance with Kaplan–Meier curves, log-rank tests, SHAP and STRING networks.

Project Intern — Jan 2024 – Jun 2024
Centre for Toxicology and Development, SRIHER — Chennai, India

- Built a video-to-data pipeline for zebrafish behaviour analysis used in drug testing: tracked fish across tanks and layers, computed speed and velocity, and produced charts and reports.

Research Intern — Dec 2022 – Feb 2023
IIITDM Kancheepuram — Chennai, India

- Developed and compared CNN breast cancer classifiers using pre-trained VGG16, ResNet, MobileNet and EfficientNet models.

Intern — Feb 2022 – Apr 2022
Smartail Pvt. Ltd. — Chennai, India

- Containerised web applications with Docker and deployed them through GitLab CI/CD pipelines.
- Worked on image processing with Detectron2, including hyperparameter tuning.

PROJECTS

Conflict Risk Monitor 2026
Python · pandas · scikit-learn · GitHub Actions · JavaScript Live demo · GitHub

- End-to-end pipeline: downloads UCDP conflict-event data (final and monthly releases) and World Bank indicators, builds a 176-country monthly panel, and publishes JSON to a static dashboard.
- Estimates the probability of 25+ battle deaths per country in the next three months; gradient boosting chosen on a validation period scored ROC AUC 0.989 and average precision 0.959 on unseen 2022–26 data.

Tamil Thedal (தமிழ் தேடல்) 2026
JavaScript · OpenCV.js · Tesseract.js · Python ETL Live demo · GitHub

- ETL script builds 1,549 verses from open sources at a pinned commit, keeps only public-domain fields, adds themes, and is rebuilt by GitHub Actions with tests.
- Search engine for Tamil literature (1,549 verses) with sound-alike matching of romanised Tamil and a TF-IDF chat helper that runs fully in the browser.

Ovarian cancer survival prediction (HGSOC) 2025
XGBoost · SHAP · TCGA GitHub

- XGBoost on TCGA clinical and mutation data with SHAP explanations; found and removed follow-up variables that leaked the outcome, then reran the notebook end to end in GitHub Actions: test accuracy 0.78, ROC AUC 0.91.

ECG heartbeat classification 2025
1D CNN · Keras GitHub

- 1D CNN classifying MIT-BIH heartbeats into five classes: 96.01% test accuracy on 21,892 held-out beats, reproduced by an automated GitHub Actions rerun.

Brain tumour classification from MRI 2025
CNN · OpenCV · Streamlit · pytest GitHub

- Four-class CNN (0.80 accuracy, 0.77 macro F1 on 2,258 test images) and ResNet50 transfer learning, deployed as a Streamlit app with unit tests and CI.

OTHER PROJECTS

- **Emotion classification from tweets** (BiLSTM · TensorFlow · Gradio, 2025): Dual-input BiLSTM combining word embeddings with sentiment polarity: 90.63% test accuracy, served as a Gradio app.
- **Handwritten prescription reader** (Detectron2 · object detection, 2023): Detectron2 model that locates handwritten medicine names in prescription images, trained on a custom annotated dataset.
- **Student engagement analysis (VLE)** (scikit-learn · ensembles, 2025): Voting ensembles (random forest, logistic regression, gradient boosting, SVM) on virtual learning environment data, compared across feature-selection settings.
- **Retail sales register analysis** (Requests · Beautiful Soup · pandas, 2025): Scraped multi-page sales records, cleaned dates, currencies and missing values, and analysed revenue, profit and customer behaviour by location and age group.
- **RNA-seq analysis of HER2 breast cancer** (R · DESeq2 · clusterProfiler, 2024): Differential expression, pathway enrichment and survival predictors in R to study how HER2 amplification changes gene expression.
- **Wimbledon Championship dashboard** (Tableau, 2021): Interactive Tableau dashboard of Wimbledon results and player statistics.
- **Teeth segmentation in dental X-rays** (U-Net, 2024): Semantic segmentation of panoramic dental X-rays with U-Net for diagnostic measurement.

AWARDS AND ACTIVITIES

- UCD Innovation Challenge 2024: Innovation Spark Award
- NASA Space Apps Challenge 2024: built a climate storytelling website with a team
- UCD Advantage Award
- First place, Tamil poetry competition (Agal Tamil Club)
- Second place, Palaestra intramural volleyball tournament